

Hihi nesting success

A spatial analysis to determine degree of nesting success and to predict number of hihi fledglings raised



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<https://www.birdoftheyear.org.nz/stitchbird/>

Table of Contents

Abstract.....	2
Introduction	2
Methodology.....	3
Data set preparation	3
Environmental data	3
Nest box and site data	4
Local nesting success analysis.....	4
k-Nearest neighbors.....	5
Getis-Ord Gi-star statistics	5
Change in hihi local hot spots map	6
Random Forest method: what it is and its suitability	6
Results and Discussion	7
Fledgling and nest box analysis.....	7
Nesting sites success maps	8
kNN maps of degree of site success per period	8
Statistics of kNN maps	10
Random Forest analysis.....	12
Random forest statistics per period	12
Random forest maps per period	14
Conclusion	15
Acknowledgements	16
A note on the supplementary data	16
Appendix	17
Forest type definitions	17
Additional statistical analysis	18

Abstract

This report explores the spatial patterns seen in the nesting data for hihi (*Notiomystis cincta*) in Zealandia Te Māra a Tāne, a fenced ecosanctuary in Wellington, New Zealand. Analyses focus on the number of fledglings raised in five periods, each ranging four to five years, starting in 2005 when hihi were reintroduced to the sanctuary. Nesting success maps and statistics using k-nearest neighbor method and Gi* statistics per period were made to visualize and study the degree of each nesting site's success. Another map was made that showed the changes in nest site occupancy and their nesting success that encompassed all five periods. Maps to predict nesting success of locations within the sanctuary along with statistics using the machine learning Random Forest method, were made for each period. The findings from these maps provide a visual record of a population crash that began around 2019 showing a drop in nesting success and lower predictions in fledglings raised for several years. The Random Forest analysis revealed that some environmental characteristics were associated with nesting success, however the degree of importance of each environmental characteristic varied substantially between each period. The maps and the findings in this report can aid biologists and conservationists in their efforts in monitoring the hihi's nesting success in Zealandia.

Introduction

Hihi, also known as stitchbird (*Notiomystis cincta*), are an endemic species historically residing throughout the North Island of New Zealand, which were extirpated from the mainland by the late 19th century. For the past 21 years since their reintroduction to the mainland at Zealandia Ecosanctuary, staff and volunteers have strived to increase the numbers of the population, with the long-term goal of making it a self-sustaining population. Nest boxes designed specifically for hihi are located throughout the park at designated sites. Each site is labeled with an ID number, and each nest box at the site is labeled with the site ID number followed by a letter. GPS coordinates for most of the nest boxes are documented. There are one to three nest boxes per site.

Since the hihi's introduction in 2005, data on which hihi female inhabited each nest box and the numbers of eggs laid, chicks hatched, and fledglings raised in each nest box per season has been collected to monitor the species' population. After a steady rise in numbers during the first ten years post-reintroduction, there was a population crash that began in 2019, and resulted in poor nesting success for the 2021 and 2022 nesting seasons.

This study examines the success of nesting sites through spatial analysis. **Success in this study is defined as the number of hihi fledglings successfully raised per nesting site, per season.**

This study has two goals:

- 1) *Create maps showing local high and low nesting success per site over the past 21 years.*

Local high and low nesting site success will be done on the site scale. Degrees of high and low success will be determined using Gi* statistics. Mapping and labeling the sites based on their degree of success will show fluctuation in success over time, as well as the degree and location of fluctuations. A further analysis on how popularity/trends of nesting sites changed over time per period will also be done. Identifying the level and location of nesting site success can help with future conservation decision making.

- 2) *Create maps showing where ideal nesting sites across the entirety of Zealandia are, based on predictions of degree of success (number of fledglings raised) and several environmental factors.*

Prediction maps will be made for five time periods that use all 21 years of hihi nesting site data as input. Such maps can aid in understanding temporal dynamics and why prediction of fledgling success for each time period varies.

The following sections include a Methodology that explains data preparation, breaks down the methods used to create the maps, and explains why these methods were chosen. The Result and Discussion sections display the maps created and tables with statistics, and discusses what the maps and statistics reveal. Lastly, the Conclusion section summarizes the findings, interpretations, and usefulness of this study.

Methodology

Data set preparation

Environmental data

Seven environmental variables were considered in this study: distance to water (m), distance to the closest point on the fence line (m), distance to nearest sugar water feeder (m), elevation (m above sea level), slope (degree), aspect (degree), and forest type (Indigenous Forest, Broadleaved Indigenous Hardwoods, or Other Exotic Forest). All datasets were collected from public sources, and more information on each data set can be found in the supplementary materials.

The only dataset found consisting of rivers and streams was collected as an image from the *ArcGIS Field Maps* app. A screen shot of this image was then georeferenced in ArcGIS Pro, and the locations of streams were traced over and saved as a new layer. Each river/stream in this dataset has a width of one meter. This new stream layer was then merged with a layer consisting of the two lakes present in the park, resulting in a layer called “water bodies”.

The water bodies layer, along with the point dataset showcasing locations of sugar water feeders, and a polygon dataset of Zealandia’s border, were used to calculate distance to the nearest water source, feeder, and fence line, respectively. These three distance calculations used the same ArcGIS Pro tool, “Near”, which calculates the distance between the site and the closest feature of interest, and results in a raster layer with this information.

Elevation, slope degree, and aspect degree were all calculated from the LiDAR Digital Elevation Model (DEM) raster layer, and resulted in raster layers with these characteristics.

After overlaying the forest type dataset with the hihi nesting site dataset, all sites are located in one of three forest types: indigenous forest, broadleaved indigenous hardwoods, or other exotic forest. For an explanation of each forest type, please refer to the Appendix. Three binary raster layers were thus created, each showing which areas of the park contained one of the forest types, and used the value of 1 (present), while other forest areas that did not contain that forest type had the value of 0 (not present).

Values of each of these environmental characteristics were then extracted from the exact location of each site, and appended as new columns to the site dataset.

Nest box and site data

Nest box data was prepared through several steps. Some nest boxes were lacking coordinates. There were also several nest boxes that had coordinates, but no nesting data; these nest boxes were excluded from the analysis.

Because not all GPS coordinates were available for each nest box, ultimately the analysis was carried forward with using nesting sites. Fledglings raised in the same site, in the same year, were summed to determine the total number of fledglings raised at a site for each nesting season. The coordinates used for the nesting site was taken from the nest box at the site that produced the most fledglings.

All environmental characteristics that were calculated at the site locations were added in as new columns:

1. Distance to the nearest water body (m)
2. Distance to the nearest part of the fence on the boundary of the sanctuary (m)
3. Type of forest site is located in (indigenous forest, broadleaved indigenous hardwoods, or exotic forest)
4. Slope degree the site is located at (degree)
5. Aspect degree of the slope the site is located at (degree)
6. Distance to the nearest feeder (m)
7. Elevation¹ (m)

This final version of the site dataset, including all the characteristics of the environmental data, was then grouped into five time periods of four years (except for the first period, which contains five years): 2005-2009, 2010-2013, 2014-2017, 2018-2021, and 2022-2025. Grouping years into periods allows for better interpretation of the analysis of temporal change in site success. The total number of sites present in each temporal period are displayed in Table 1.

Table 1: Number of active sites for each period.

Period	n
2005-2009	45
2010-2013	71
2014-2017	98
2018-2021	79
2022-2025	68

Local nesting success analysis

The nesting success per site analysis was done using k-nearest neighbors (kNN) and Getis-Ord Gi-star (Gi*) statistics. k-nearest neighbors defines spatial neighborhoods of nesting sites, and Gi* detects clusters of high or low predicted success. The resulting maps then show where those clusters occur.

¹ The elevation of the ground, therefore excluding the additional height of the placement of the nest box on the tree. All nest boxes are located about 1.5 meters up the tree.

k-Nearest neighbors

k-nearest neighbor (kNN) is an algorithm in spatial analysis and machine learning. For each point, or nesting site, it finds the closest other specified number of points (sites) based on distance. In this study the specified number was five. These close points are “neighbors”.

The kNN method was chosen because of the arrangement of the physical locations of the sites. Distances between sites are uneven, and the kNN method is suitable for handling such uneven physical arrangement, than for example a fixed distance band of 100 or 150 meters. kNN gives every site the same number of neighbors (in this case five), regardless of spacing, when determining degree of success. As this analysis finds the degree of success using G_i^* statistics, kNN also ensures that there are no empty neighbor sets, there are equal neighborhood sizes (allowing for comparable Z-scores), and there are no artificial high or low nesting success caused by isolated sites. Isolated sites are also not unfairly labelled as areas of low nesting success.

kNN is also an ideal method for this study because the nest boxes were installed by humans, meaning that, to an extent, placement is not random. Additionally, exposure to environmental variables occur locally around the nest boxes and sites, and kNN is able to capture this local spatial structure/characteristics, without forcing an arbitrary radius as a fixed distance method might.

Using five nearest neighbors was chosen for several reasons. Using a small number is better for statistical stability of the model. Smaller amounts of neighbors than five could introduce noise to the G_i^* statistics, and areas of nesting success may appear more random, with results varying strongly between periods. On the other hand, using larger number of nearest neighbors can cause local hot spots to be smoothed out, and could cause averaging of result values.

Five hot spot maps were created, one for each time period using the kNN analysis. Each maps label each site during that period as a local hot spot or cold spot (along with the degree), or as not significant.

Getis-Ord G_i^* statistics

G_i^* statistics were calculated, measuring spatial clustering strength and showing where high or low values cluster in space more than expected by chance.

G_i^* are z-scores:

- 0 = no spatial clustering
- +1.65 = hotspot at 90% confidence
- +1.96 = hotspot at 95% confidence
- +2.58 = hotspot at 99% confidence
- -1.65 / -1.96 / -2.58 = cold spots at the same confidence levels

These values represent a nesting site in a cluster of a certain predicted success, with x standard deviations above what a random chance would produce. It reveals where high-success or low-success sites cluster in space.

Non-significant values mean that there is no strong spatial structure present, and that the nesting site is not part of a high or low area of nesting success. A site attains this value when the surrounding sites have success values that look random or mixed, not forming a pattern.

Change in hihi local hot spots map

Another map was made that showed which sites deemed as local hot spots changed throughout the five periods. This map was made by combining the Getis-Ord Gi* analyses of all the five period maps showing degrees of local nesting success. A Gi_Bin field analysis was run where sites classified as 90% confidence hot spots were given the value 1, 95% confidence hot spots were given the value of 2, and 99% confidence hot spots were given the value 3. Sites not classified as any of these were given the value of 0. The values were then summed together, producing the change in local hot spots map.

Random Forest method: what it is and its suitability

A Random Forest model was run in R to predict fledgling success based on the environmental characteristics of the site's location. The number of "trees" used was 500. See figure 1 for a visualization for how a Random Forest model works. A Random Forest (RF) is a machine-learning method that builds hundreds of decision trees, where each tree learns different ecological rules. Each tree has a random subset of sites and environmental variables, and learns its own set of ecological rules. The final result, the outcome prediction, is the average predictions from all trees. This makes the RF model stable, robust, resistant to overfitting, and able to model complex ecological relationships. RF is suitable for this study because of the dataset's nonlinear relationships, interactions between variables, mixed data types (continuous + categorical), spatial autocorrelation, ecological noise, temporal instability, and missing or imperfect predictors.

An RF model was run for each of the five time periods. Each of these prediction maps used the nesting data from that time period, as well as the environmental values calculated for each site's location. This resulted in five maps that predicted fledgling success throughout the park for each period.

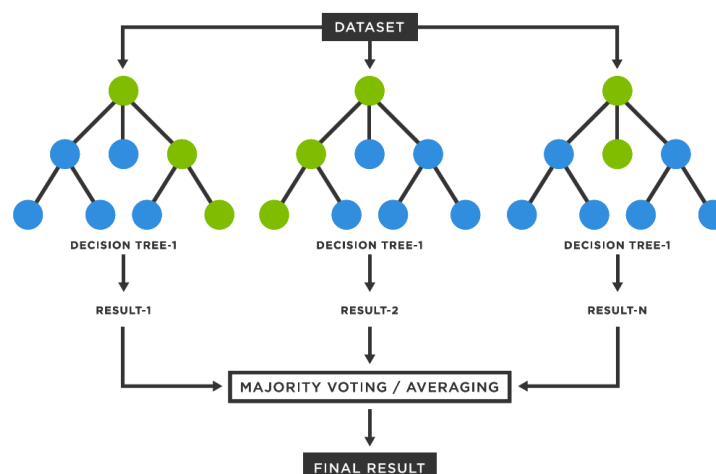


Figure 1: Depiction of how a Random Forest model works. Source: <https://medium.com/@denizgunay/random-forest-af5bde5d7e1e>.

Results and Discussion

Fledgling and nest box analysis

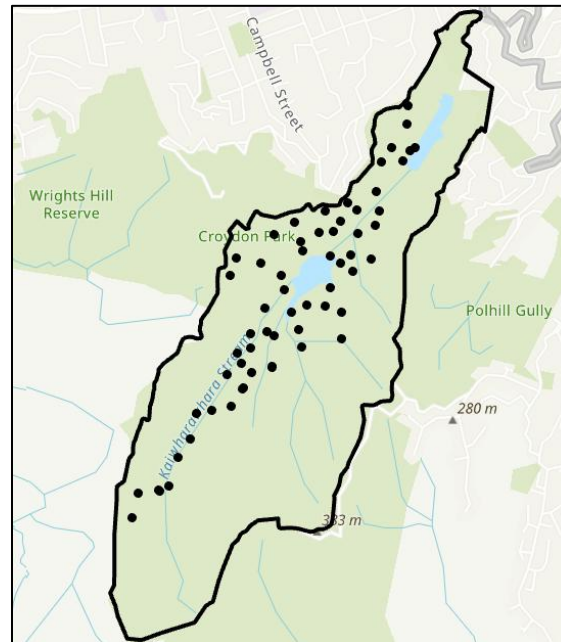


Figure 2: Location of all hihi nesting sites throughout Zealandia as of 2025.

Table 2: Summary statistics table of the distribution of fledgling counts per period. Values are fledglings per site.

period	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	SD
2005-2009	0.00	0	3	3.02	5	9	2.68
2010-2013	0.00	2	3	3.37	5	8	2.12
2014-2017	0.00	0	2	2.46	4	9	2.22
2018-2021	0.00	0	2	2.13	3.5	8	1.89
2022-2025	0.00	0	3	2.84	4.25	8	2.48

Table 2 shows fledgling statistics per site, per period. The numbers of fledglings raised ranges from zero to eight or nine for each period, and the average number of fledglings raised was always between 2 and 3. This is in line with the RF maps, as many areas in the park are also predicted to raise 2 to 3 fledglings.

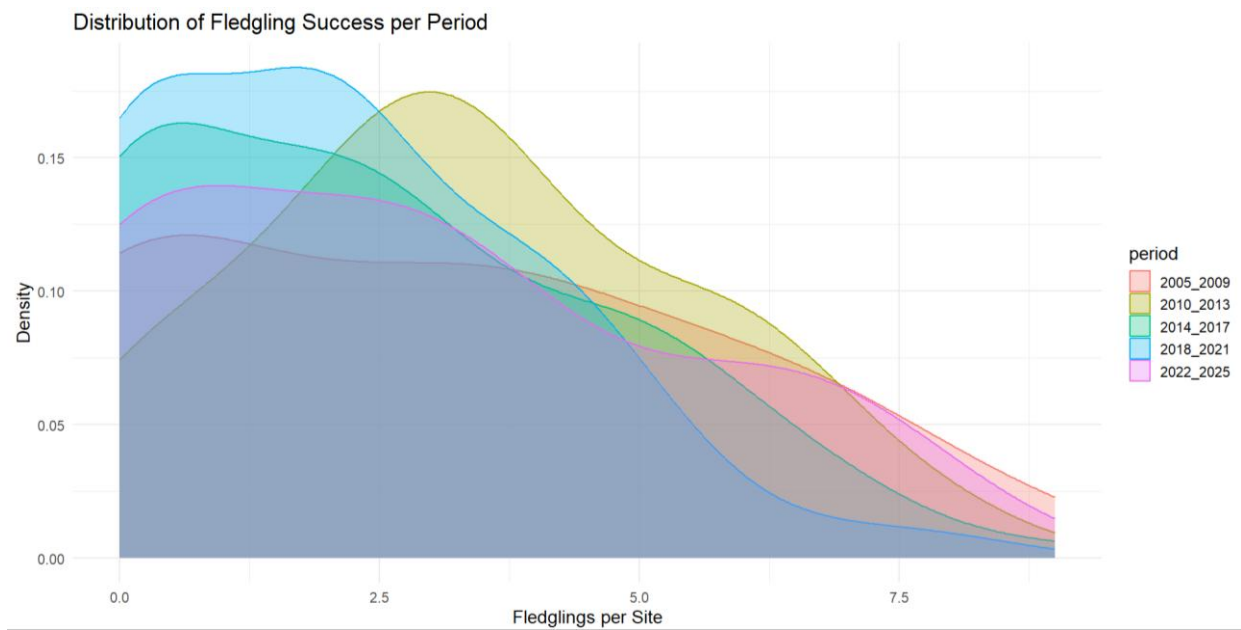


Figure 3: This figure shows the distribution of fledgling success per period. The x axis shows the possible number of fledglings per site, and the y axis shows the density, or how concentrated the data is around each fledgling success value.

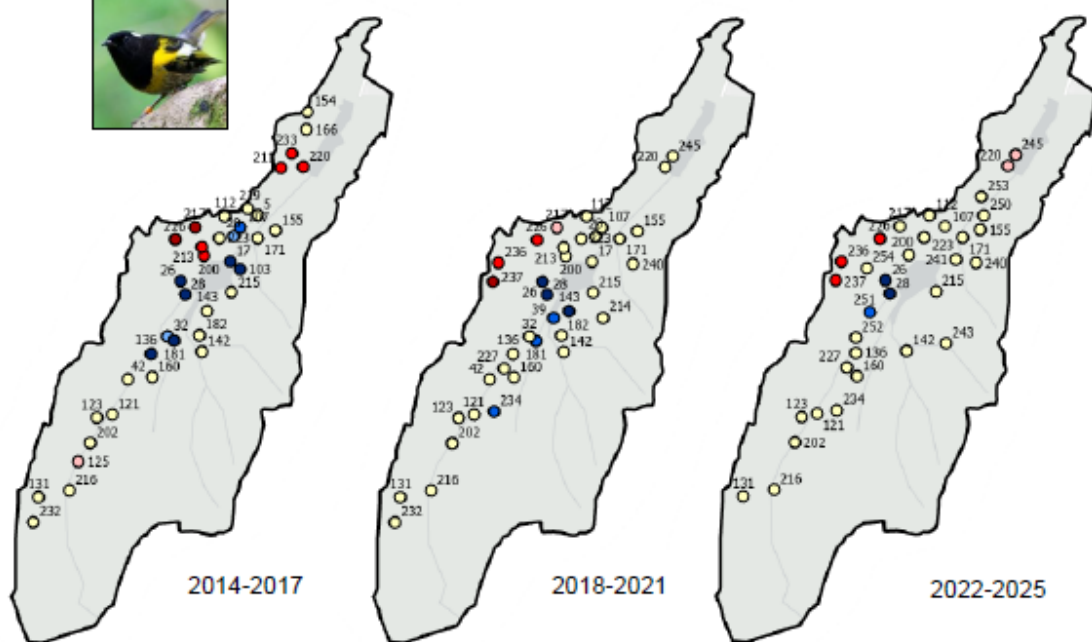
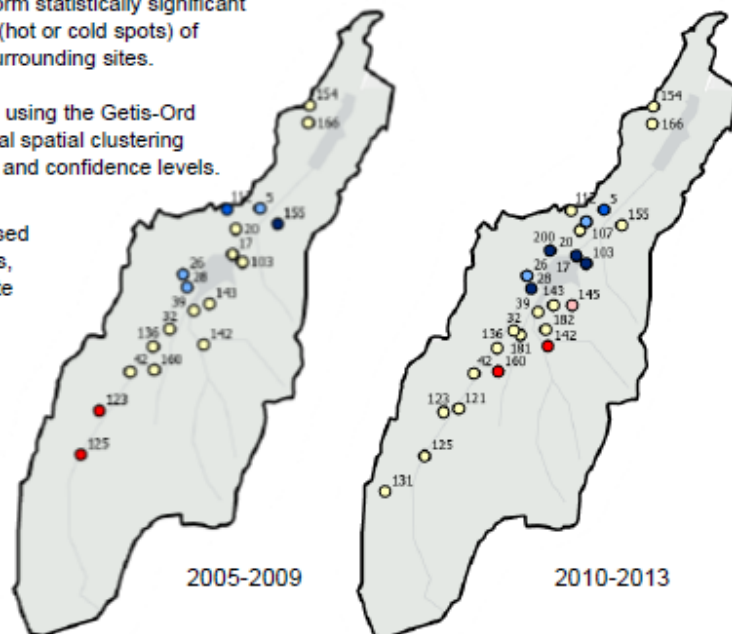
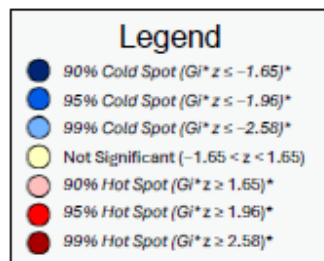
Nesting sites success maps

kNN maps of degree of site success per period

Hihi local nesting success per site in Zealandia for five time periods



- These maps show which sites form statistically significant clusters of high or low numbers (hot or cold spots) of fledglings raised compared to surrounding sites.
- Hot and cold spots are identified using the Getis-Ord G_i^* statistic, which evaluates local spatial clustering based on standardized z-scores and confidence levels.
- Each map displays fledgling-based site success for five time periods, with each site labeled with its site number.



0 0.25 0.5 1 Kilometers

Figure 4: Maps showing the success of each site per period.

Figure 4 labels each nesting site as an area of high/low success (and its magnitude), or as non-significant. There tends to be clustering of local hot spots (multiple sites with high success), local cold spots (multiple site with low success), and non-significant spots in all the maps.

Statistics of kNN maps

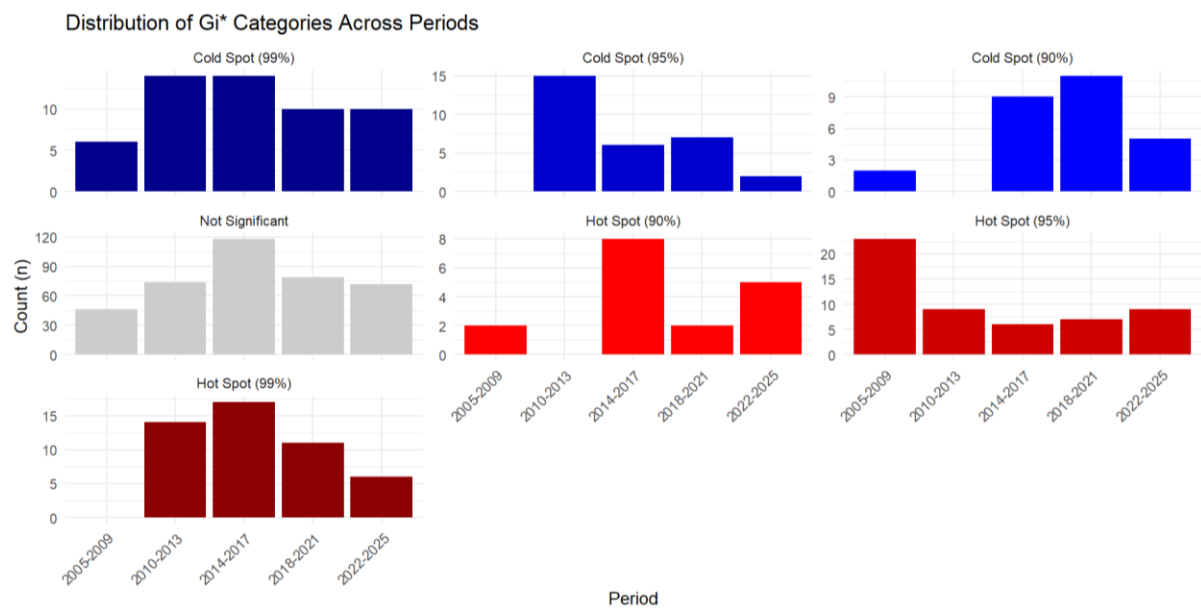


Figure 5: Distribution of number of sites per period falling in each G_i^* category. All bars representing periods are displayed in chronological order. For a table version of this data, please refer to the Appendix.

Change in local nesting success across periods per site

Change in hihi local hot spots 2005-2025

How the success of the number of fledglings raised per site has changed throughout the years

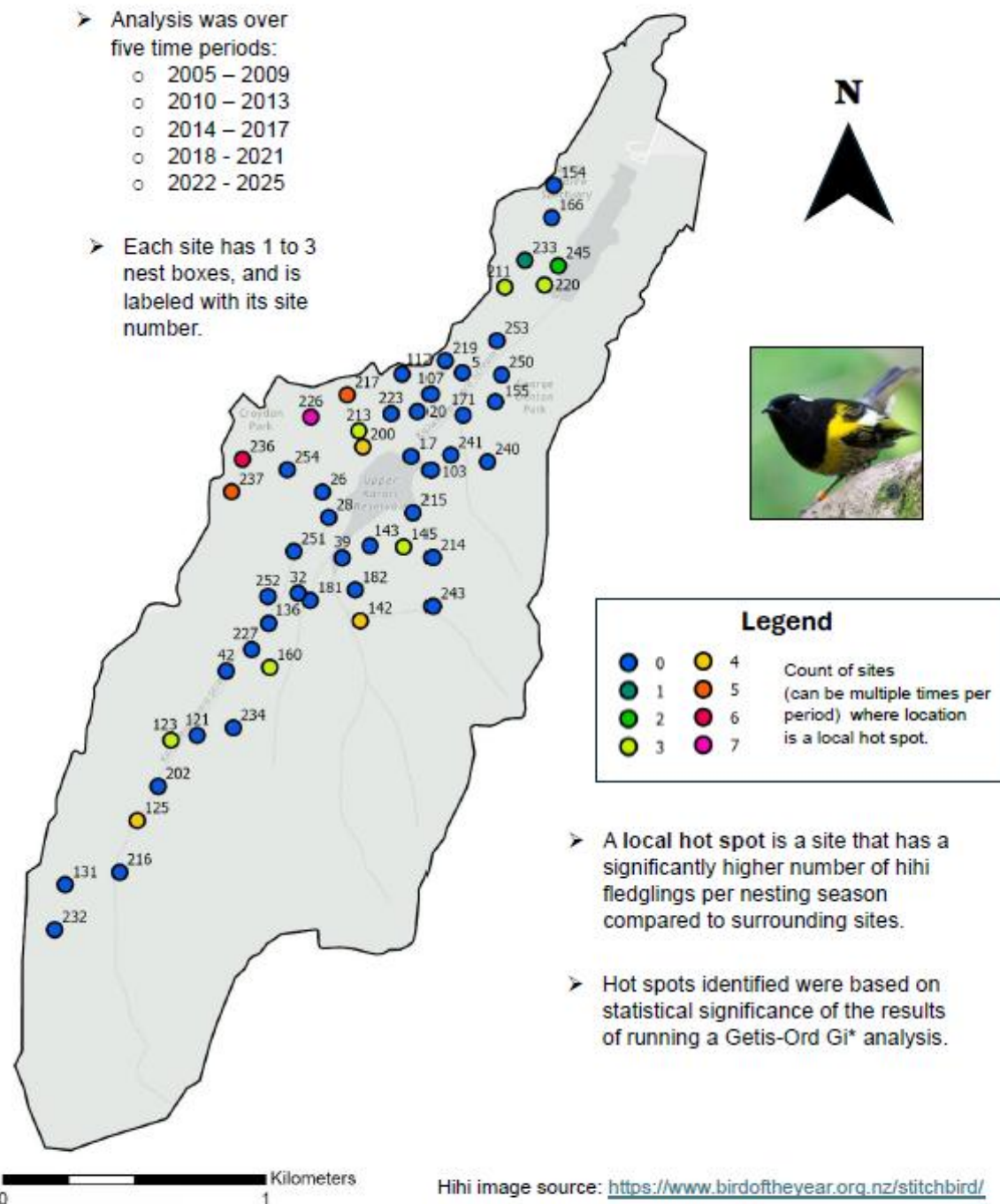


Figure 6: Change in nesting success at sites across periods.

Figure 6 shows the temporal behavior and stability of each nesting site's success across all periods. Sites with the value of zero were never part of a statistically significant hot spot in any of the five periods, and never performed well. Sites with lower counts were unreliable nesting areas. Sites with higher counts were nesting areas with long-term mid to high success.

Table 3: local hot spot intensity across periods.

Period	Mean Gi*	Max Gi*	Min Gi*
2005-2009	-0.61	2.04	-2.88
2010-2013	-0.22	2.21	-3.41
2014-2017	-0.13	2.82	-3.78
2018-2021	-0.13	2.59	-5.06
2022-2025	-0.50	2.21	-5.87

Table 3 shows summaries of Getis-Ord Gi-star (Gi*) values for each time period. Gi* values are standardized z-scores and are a statistical measure of spatial clustering. The more positive the value, the higher the fledgling success, creating a "hot spot" for a site. The mean and max values show the strongest hot/cold spots in that period. The mean Gi* values show the overall tendency toward hot or cold spots. Looking at table 3, while nest success was spatially structured, local cold spots were stronger in magnitude than local hot spots.

Random Forest analysis and prediction maps

Random forest statistics per period

Table 4: The Mean Squared Error (MSE) and R squared values for each period.

Period	MSE	R squared
2005-2009	6.61	0.06
2010-2013	5.30	-0.20
2014-2017	6.06	-0.24
2018-2021	5.68	-0.61
2022-2025	6.54	-0.08

The *Mean Squared Error* (MSE) measures the average squared difference between estimated values and the actual true values. It evaluates the accuracy of a predictive model or estimator, where a lower MSE signifies a better fit and more precise predictions. Almost all periods show negative R^2 , meaning that the predictors together do not explain nest success. However, Table 5 (below) shows that considering predictors separately shows a fluctuation in its importance to a site's success.

Table 5: Variable importance measure: Percent increase in MSE when a variable is randomized.

Environmental variable	2005-2009	2010-2013	2014-2017	2018-2021	2022-2025
Slope degree	5.22	-0.96	-1.61	-9.38	10.56
Aspect degree	4.32	3.07	4.56	-8.06	7.05
Water distance	7.90	0.82	1.13	-10.35	3.78
Border distance	9.05	6.09	2.54	-8.75	2.73
Elevation	7.40	1.42	4.73	-10.15	-1.40
Feeder distance	3.77	-0.19	4.49	-8.83	-1.00
Forest broadleaf	-0.24	-0.82	4.54	-3.31	1.14
Forest exotic	0.00	-2.67	-1.54	-3.91	-1.65
Forest indigenous	2.10	-1.00	1.69	-2.16	1.8

Table 5 is a measure of the importance of each environmental variable. It shows how much worse the Random Forest model predicts fledgling success when that variable is not included. Therefore, high values indicate this environmental variable matters, and is a useful predictor of fledgling success in this period. Low values indicate that the variable adds little predictive value to the model. Negative values indicate that the variable gives noise to the model, and that the model predicts better without it. This table shows that the degree of importance of each environmental variable varies drastically between periods, and the lack of a pattern leads to the negative R squared values seen in Table 4.

It is important to note that Table 5 shows the importance of the environmental variable, and not the actual ideal unit of that variable. For example, slope degree had a value of 10.56 in 2022-2025. This means that slope degree was a significant factor for fledgling success in this period, but does not indicate which range(s) of the slope's degree was ideal in this period.

Briefly, looking at the results in Table 5, 2005-2009 shows strong habitat-driven success. In 2010-2013, the habitat signal weakens, but the signal returns in 2014-2017. In 2018-2021, the habitat becomes noise, meaning that success is driven by unmeasured factors. In 2022-2025, the habitat signal partially recovers. Therefore, habitat with certain environmental characteristics mattered, and quite strongly during some periods, but which environmental characteristics mattered, as well as their level of importance, changed between periods.

Prediction maps per period

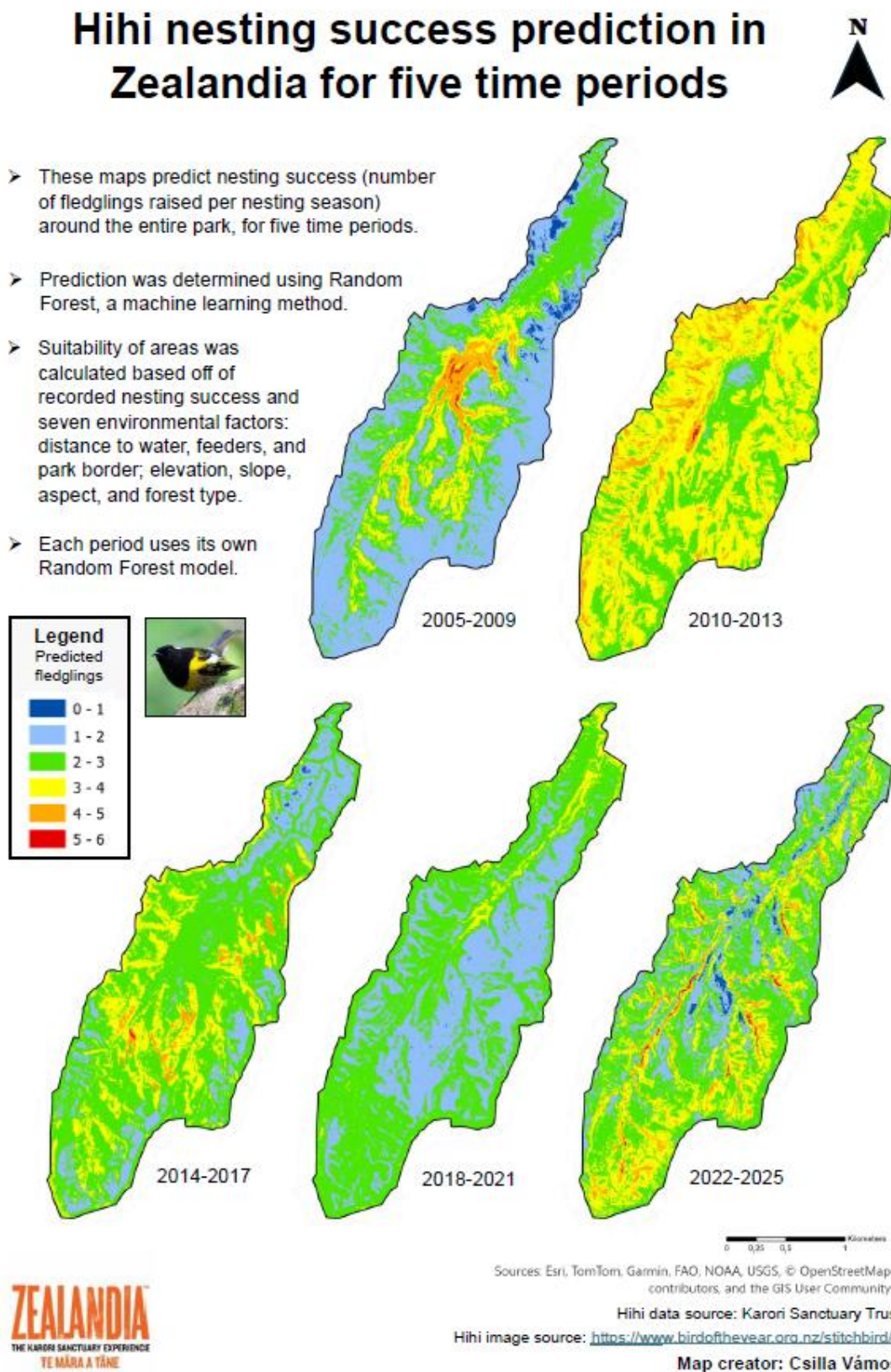


Figure 7: Prediction maps for each time period created using Random Forest models. Scale is the number of fledglings predicted to be successfully raised should a site exist in that exact location.

Figure 7 shows prediction maps for each period, with a resolution of 1 x 1 m, meaning that every square meter in the park was queried to determine the number of predicted fledglings per site, if a nesting site existed in that exact 1 x 1 m area. These maps were created using Random Forest models, and considered the seven environmental factors in their prediction of fledgling success.

The period 2005-2009 has one large concentrated area of high fledgling numbers, this is because hihis were introduced to Zealandia in 2005 and thus it took a few years for the population to settle in. In all the period maps, there are areas that have a variety of colors within a small amount of area, indicating a variety of success in those areas. Other areas have large swaths of the same color, indicating stable prediction numbers throughout an area. It can be seen that there are already indications of a pre-crash in population in the 2014-2017 map, before the actual population crash happens in the 2018-2021 map. The following period, 2022-2025, then more closely resembles the growth period seen in 2010-2013, when population numbers were robust.

The prediction maps look very different from period to period because each period has its own Random Forest model, and each model has differences in training data, environmental relationships, variable importance, ecological conditions, and fledgling success distributions. The tables of the statistics of the Random Forest analyses from the previous section (Table 5) support the observation that fledgling success changed dramatically across time. This is because some periods are driven by different magnitudes of each of the environmental variables. Some variables in some periods contribute to more noise than others. Therefore, the different patterns in each map is expected.

These prediction maps are useful for various reasons:

- Detecting ecological change
- Understanding temporal dynamics
- Identifying persistent hotspots
- Identifying transient hotspots
- Understanding why prediction in success varies
- Interpreting environmental drivers
- Conservation decision-making

Conclusion

The local nesting success maps using the kNN method show the locations of local hot, cold, and non-significant sites per period, and their clustering in space. The change in local hot spots map provides an overview of the success of each nest box when considering all periods. These maps are useful for examining the success of nesting sites and monitoring how this may have changed over time.

The Random Forest/Prediction period maps and their statistics are useful to examine how areas around the park that were considered suitable areas to foster fledgling success changed over time when considering several environmental variables. The prediction maps and their statistics from 2005-2009 and 2010-2013 indicate that nesting success is habitat-driven and predictable. The periods 2014-2017 and 2018-2021 show ecological instability,

which is in line with the known population crash that started around 2019, and these maps and statistics suggest signs to the population crash already beginning to happen in the previous period of 2024-2017. The statistics and maps for the 2022-2025 period show partial recovery in habitat driven nesting success.

In regards to the population crash, it is not known what drove it. While it could have been food availability shifts, climate, nest box competition, unmeasured ecological drivers, or something else, examining what caused the population crash is out of scope for this study.

The produced maps are useful for examining the location and degree of a nesting site's success, and how this changed over time as well as drivers in success, and can shed light on variables that can boost a nesting site's success.

Acknowledgements

I would like to thank Jo Ledington for organizing a permit to access the nesting data, to Zealandia for allowing me to carry out a spatial analysis on their data, and to Tas Vámos for helping prepare the research contract with Zealandia and for his feedback on the maps and this report.

A note on the supplementary data

The digital folder of supplementary materials associated with this report include the following:

- excel file of meta data on all environmental data used
- all the environmental data used, raw and transformed
- PDF of the local hot spot maps per period
- PDF of the change in hot spots throughout the periods
- PDF of the prediction maps per period
- CSV of the hihi nesting data that includes the values of the environmental characteristics extracted for each nest box location.

Appendix

Forest type definitions

Below are excerpts from the document: *New Zealand Land Cover Database Version 2: Illustrated Guide to Target Classes* (Thompson et al., 2003)². These excerpts explain what each forest class is.

Indigenous Forest (page 104):

Classification of indigenous forest classes using satellite imagery has not been attempted for LCDB2. A national programme establishing permanent plots throughout all indigenous forest and scrub mapped in LCDB1, began in 2002. Ground data from this programme and other survey records may be used to update indigenous forest classes at a later date. Indigenous forest is defined as vegetation dominated by indigenous tall forest canopy species.

Broadleaved Indigenous Hardwoods (page 74):

Typically found in high rainfall areas associated with Low Producing Exotic Grassland in hill country throughout New Zealand. However, the class also includes low-growing, coastal broadleaved forest. Characteristic is the presence of a mix of broad-leaved, generally seral hardwood species, such as wineberry (*Aristotelia serrata*), mahoe (*Melicytus ramiflorus*), *Pseudopanax* spp., *Pittosporum* spp., *Fuchsia* spp., ngaio (*Myoporum laetum*), and titoki (*Alectryon excelsus*), together with tutu (*Coriaria* spp.) and tree ferns. The presence of this class usually indicates an advanced successional stage back to indigenous forest. Canopy height ranges from 3 - 10m.

Other Exotic Forest (page 98):

Exotic forest consisting of conifers other than *Pinus radiata*, such as Douglas fir, macrocarpa, and larch, or evergreen broad-leaved species, such as Acacia and Eucalyptus. This class also includes stands of wilding pines (i.e. usually Corsican, Contorta and Ponderosa Pine), as far as they are identifiable in the satellite imagery, i.e. stands are dense/tall enough to show a distinct signature.

² Thompson, S., Grüner, I., & Gapare, N. (2003). *New Zealand land cover database version 2: Illustrated guide to target classes*. Ministry for the Environment.

<https://www.tupu.nz/media/tnpbnbxx/new-zealand-land-cover-database-version-2-illustrated-guide-to-target-classes.pdf>

Additional statistical analysis

Table 1A: variable importance per time period: Increase in Node Purity.

Environmental variable	2005-2009	2010-2013	2014-2017	2018-2021	2022-2025
Slope degree	9.66	9.65	19.19	11.93	29.01
Aspect degree	10.53	21.76	24.70	12.93	33.61
Water distance	13.41	17.64	22.27	19.08	20.57
Border distance	12.97	20.35	22.93	16.01	21.00
Elevation	19.13	12.45	23.90	19.64	20.72
Feeder distance	19.25	12.50	25.66	16.16	13.12
Forest broadleaf	1.40	2.32	4.78	1.48	1.85
Forest exotic	0.00	2.01	6.72	2.05	0.56
Forest indigenous	1.18	1.29	3.03	1.36	2.57

Table 1A's increase in Node Purity show how much each variable helps split in the data inside the Random Forest. It is based on the total reduction of residual sum of squares, contributed by each variable across all "trees". Therefore, this measurement shows how much a variable reduces prediction error within the "trees".

Table 2A: hot spot summary statistics per period

Period	Gi* Category	n
2005-2009	Cold Spot (90%)	2
2005-2009	Cold Spot (99%)	6
2005-2009	Hot Spot (90%)	2
2005-2009	Hot Spot (95%)	23
2005-2009	Not Significant	46
2010-2013	Cold Spot (95%)	15
2010-2013	Cold Spot (99%)	14
2010-2013	Hot Spot (95%)	9
2010-2013	Hot Spot (99%)	14
2010-2013	Not Significant	74
2014-2017	Cold Spot (90%)	9
2014-2017	Cold Spot (95%)	6
2014-2017	Cold Spot (99%)	14
2014-2017	Hot Spot (90%)	8
2014-2017	Hot Spot (95%)	6
2014-2017	Hot Spot (99%)	17
2014-2017	Not Significant	118
2018-2021	Cold Spot (90%)	11
2018-2021	Cold Spot (95%)	7
2018-2021	Cold Spot (99%)	10
2018-2021	Hot Spot (90%)	2
2018-2021	Hot Spot (95%)	7
2018-2021	Hot Spot (99%)	11
2018-2021	Not Significant	79
2022-2025	Cold Spot (90%)	5

2022-2025	Cold Spot (95%)	2
2022-2025	Cold Spot (99%)	10
2022-2025	Hot Spot (90%)	5
2022-2025	Hot Spot (95%)	9
2022-2025	Hot Spot (99%)	6
2022-2025	Not Significant	72